Sankit Sharma

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.60
Ragendra Swarup Public School	Class XII (ISC)	2017	73.00%
Holy Public Junior College	Class X (CBSE)	2015	9.00%

WORK EXPERIENCE

Senior Engineer | Mechanical BIM Dept., Pinnacle Infotech Solutions

- Collaborated with cross-functional teams and coordinated with US clients to manage project execution, communication, and timely delivery in a fast-paced professional environment.
- Managed a US-based project independently and led HVAC service coordination for DEAP projects.
- Utilized Revit, Fusion 360, SW and Inventor for BIM modelling, design coordination, and project execution across mechanical engineering projects.

PROJECTS

Analysis & Optimization of End-Effector

- Performed nonlinear static & fatigue FEA of a press-fit pin-joint gripper in ANSYS, resolving a mesh-dependent contact singularity via stress linearization and validating results against analytical (Lamé) calculations.
- Ran S-N based fatigue analysis and parametric design studies (interference, pin diameter) to quantify their effect on joint fatigue life and safety factor.

Prediction of geometric correction factor of a cracked plate using FEA & ML model

- Performed a **parametric nonlinear FEA** study on a centre- cracked stainless- steel plate and applied stress to extract SIF data. Built an ML pipeline using **polynomial regression** on the ABAQUS generated dataset to predict the geometric correction factor, to get SIF.
- Validated results by comparing Analytical, FEA, and ML-predicted values, achieving **deviation within 3.5%** across all methods.

Prediction of Elastic Modulus & Ultimate Tensile Strength of Iron Alloys using ML

- Implemented and compared three regression models -- Linear Regression, Random Forest and Gradient Boosting using data preprocessing, feature scaling and MSE, RMSE and R^2-based evaluation.
- Achieved R^2 values of 0.9203- 0.9551 , with **Random Forest attaining the highest R^2 of 0.9551** , demonstrating strong capability for data-driven prediction of alloy mechanical properties.

Classify Engine Operational Mode and Forecast Fuel Efficiency using ML

- Implemented a **Random Forest classification** 80:20 train- validation split, to classify engine operating modes (Idle, Cruising, Heavy Load) and forecast fuel efficiency Achieved **99% training and 31% validation accuracy** for operating- mode classification, while fuel- efficiency forecasting **yielded 4.13 MAE and 4.72 RMSE** on validation data, highlighting the impact of temporal drift and feature overlap on model generalization.

Design & Fabrication of Acoustic Impedance tube

- Designed and fabricated an acoustic impedance tube for measuring material sound absorption characteristics using a two-microphone configuration.
- Developed the experimental methodology & tested 4 materials across 5 sample configurations, achieved a 370-3000 Hz measurement range and characterized frequency-dependent absorption coefficient.

SKILLS

- **FEA:** ANSYS & ABAQUS
- **Additional:** GD&T, UMAT & MATLAB
- **Analysis:** Structural (Linear and Nonlinear), Buckling and Modal Analysis, Transient Analysis

CERTIFICATIONS

Introduction to Mechanical Engineering Design and Manufacturing with Fusion 360 | Autodesk, Coursera

Demonstrated knowledge and skills in Fusion 360 to apply in design and manufacturing workflows to take digital parts to physical prototypes.

Engineering Drawing and Computer Graphics | IITKGP, NPTEL

12 weeks course about the engineering and machine drawings and CAD model development.

POSITIONS OF RESPONSIBILITY

Alumni Coordinator | Alumni Cell, IITH

Managing the data of Alumni, organizing the Alumni meet, events and building the bridge between students and Alumni of the specific industry or profile. Recently, coordinated in the Alumni meet held on Sept 12, 2026.

Team Lead | B.tech major project, SKIT

Planned and coordinated project execution, including task allocation and scheduling for a 4-member team, ensuring timely completion.

Volunteer | 2nd ICONRER International Conference, SKIT

Managed the guests arrivals, events coordination in this international conference held in SKIT.

ACHIEVEMENTS

1. **Silver Medallist**, for scoring 84% in "Engineering Drawing and Computer Graphics", course.
2. **NPTEL Believers,** for appearing and passing the 5 courses consecutively with the score more than 60%.

RELEVANT COURSES

- Advanced Mechanics of Solids, Fracture Mechanics, Finite Element Methods and Advanced Finite Element Methods
- Machine Learning and its Applications, Soft Robotics and Topology Optimization

HOBBIES & AREA OF INTERESTS

Cricket · Badminton · Melophile · Astrophile · Railfan · Cooking